ALI HASSAN

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EDUCATION

National University of Computer and Emerging Sciences (FAST NUCES) 2023 – Present
BS Computer Science (CGPA: 3.51/4.00) Lahore, PK

Punjab College 2021 – 2023
FSc Pre-Engineering (Grade: A) RYK, PK

SKILLS

AI/ML: Multimodal RAG, Temporal Analytics, Transformers, LLM Orchestration (LangChain)
Technologies & Web: React.js, Node.js, Redux, Sockets, Express.js
Databases & Tools: MongoDB, MySQL, PostgreSQL, Firebase, Docker, Jenkins, AWS, Git/GitHub
Languages: Python, JavaScript, TypeScript, C++, Java

EXPERIENCE

Teaching Assistant Jan 2026 – Present
FAST NUCES *Lahore, PK*

- Guiding students in AWS cloud concepts including VM deployment, storage, and serverless systems.

Web Developer Intern Jun–Aug 2025
XpertSpot Systems *RYK, PK*

- Developed full-stack web applications using the MERN stack, enhancing user interface responsiveness.

Web Developer Intern Jun–Aug 2024
Graphic Cell

- Developed modular React components to build functional and interactive web interfaces.

PROJECTS

PitchVision: Tactical Strategy Recognition — *PyTorch, YOLOv8, Transformers, SoccerNet*

- Developed a spatio-temporal vision system to classify tactical strategies using large-scale player tracking data.
- Implemented a temporal transformer architecture to model engineered spatial features across match sequences.

PredictBit: Bitcoin Price Forecasting — *Python, PyTorch, BERT, LSTM, XGBoost*

- Engineered a multimodal pipeline fusing BERT-based sentiment analysis with OHLCV market data.
- Benchmarked LSTM and CNN-LSTM architectures, achieving 74.4% accuracy in price direction prediction.

Smart Interviewer: AI Assessment Engine — *Python, RAG, Speech Analysis, Embeddings*

- Developed an adaptive interview system using RAG for context-aware technical question generation.
- Integrated semantic similarity and acoustic feature extraction for real-time candidate evaluation and scoring.

DocFlow: Multimodal RAG Pipeline — *Python, LangChain, ChromaDB, LLMs, Vector Search*

- Architected a hybrid retrieval system (BM25 + Semantic Search) with cross-encoder reranking for precision.
- Integrated citation-backed reasoning using ChromaDB, ensuring grounded LLM responses.

ACHIEVEMENTS

- Dean's List of Honors (3x), FAST NUCES.
- **Certifications:** Web Development (XpertSpot, Graphic Cell), Jenkins & Linux (KodeKloud).